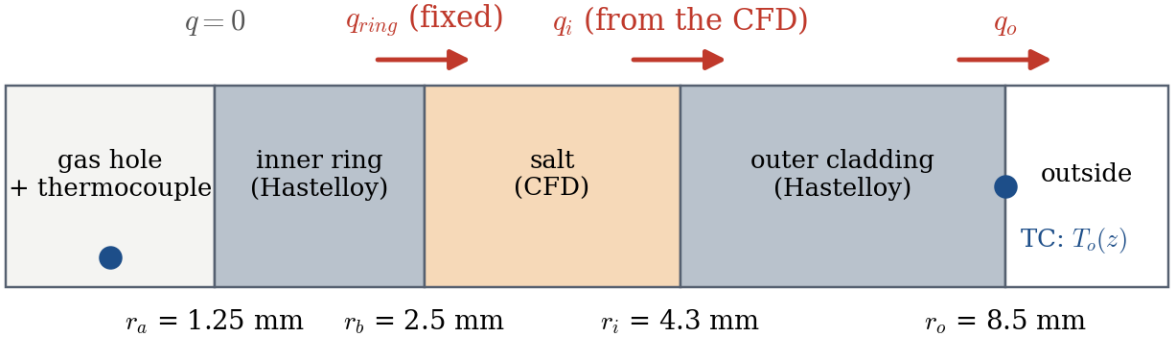
Cladding and ring boundary conditions: derivation

heatedPool, OpenFOAM v2606: how the temperature conditions on the salt walls follow from heat conduction in the Hastelloy, step by step, and how the outer-cladding and inner-ring conditions differ.

Case A, radial cut (not to scale): heat flows outward, from the ring through the salt to the outside



1. Assumptions

Steady state (the metal's thermal time constant is ~4 s); conduction in the metal only in the radial direction (axial conduction is ≤ 3–4 % of the radial heat in the outer cladding and ≤ 6–10 % in the ring); axisymmetry; perfect salt–metal contact; uniform heat generation in the metal, q_c = 2 W/g × 8.86 g/cm³ = 17.7 W/cm³; within the thin wall k is taken constant (in the code: k(T) from the Hastelloy table at the mean wall temperature).

2. The temperature equation in a tube and its integration

Energy balance on a thin ring of metal between r and r + dr (per unit height): heat conducted in minus heat conducted out plus heat generated is zero at steady state. In the limit dr → 0 this gives the heat-conduction equation in cylindrical coordinates with only radial dependence:

$$\frac{1}{r}\frac{d}{dr}\left(k\,r\frac{dT}{dr}\right) + q_c = 0 \tag{1}$$

Step 1: multiply by r, which leaves a single derivative on the left:

$$\frac{d}{dr}\left(k\,r\frac{dT}{dr}\right) = -\,q_c\,r \tag{2}$$

Step 2: integrate once in r. A is the first integration constant:

$$k\,r\frac{dT}{dr} = -\frac{q_c\,r^2}{2} + A \tag{3}$$

Step 3: divide by k r to isolate the temperature gradient:

$$\frac{dT}{dr} = -\frac{q_c\,r}{2k} + \frac{A}{k\,r} \tag{4}$$

Step 4: integrate again (the integral of 1/r is ln r). B is the second constant:

$$T(r) = -\frac{q_c\,r^2}{4k} + \frac{A}{k}\ln r + B \tag{5}$$

The radial heat flux follows from Fourier's law (positive in the +r direction, i.e. outward):

$$q(r) = -\,k\frac{dT}{dr} = \frac{q_c\,r}{2} - \frac{A}{r} \tag{6}$$

Multiplying by 2πr shows what A means: 2πr q(r) = πr² q_c − 2πA, i.e. the heat crossing the circle of radius r (per metre of height) equals the heat generated inside it, πr² q_c, minus 2πA. A is fixed by

what happens at the inner surface. **Two constants (A, B) need exactly two conditions**: which two conditions we know is the whole difference between the outer cladding and the inner ring.

3. Outer cladding (mode cladding)

Radii: salt side r_i (4.3 mm in A, 3.5 mm in B), outer surface r_o = 8.5 mm where the external thermocouples are. The two conditions:

$$q(r_i) = q_i \qquad (\text{heat flux leaving the salt}: \text{ unknown, set by the CFD})$$

$$T(r_o) = T_o(z) \qquad (\text{measured, spline through the thermocouples})$$

Constant A from the flux condition, using (6) at r = r_i:

$$q_i = \frac{q_c\, r_i}{2} - \frac{A}{r_i} \quad \Rightarrow \quad A = \frac{q_c\, r_i^2}{2} - q_i\, r_i \tag{7}$$

Temperature difference across the wall: write (5) at r_i and at r_o and subtract; B cancels:

$$T(r_i) - T(r_o) = \frac{q_c}{4k}\left(r_o^2 - r_i^2\right) - \frac{A}{k}\ln\frac{r_o}{r_i} \tag{8}$$

Substitute A and collect the terms in q_i and in q_c:

$$T_w - T_o = \frac{q_i\, r_i \ln(r_o/r_i)}{k} + \frac{q_c}{4k}\left[r_o^2 - r_i^2 - 2\, r_i^2 \ln\frac{r_o}{r_i}\right] \tag{9}$$

The two terms have a clear meaning: the first is the drop needed to conduct the **salt's heat** q_i through the wall; the second, ΔT_{gen}, is the extra drop needed to get the **cladding's own heat** out. It does not depend on q_i, so it only shifts the reference temperature.

With the cladding conductance h_c this is a Robin (mixed) law between the flux leaving the salt and the wall temperature:

$$h_c = \frac{k}{r_i \ln(r_o/r_i)} \qquad\qquad q_i = h_c\left(T_w - T_o - \Delta T_{gen}\right) \tag{10}$$

Energy check: per metre of height the outer surface rejects the salt's heat plus the cladding's own heat,

$$q_o\, r_o = q_i\, r_i + \frac{q_c}{2}\left(r_o^2 - r_i^2\right) \tag{11}$$

Numbers (k = 22.6 W/m·K): case A, h_c = 7713 W/m²K and ΔT_{gen} = 5.6 K; case B, h_c = 7277 W/m²K and ΔT_{gen} = 7.5 K. With q_i ≈ 140–175 kW/m² the salt-side wall sits ≈ 25–32 K above the outer thermocouple.

Bottom plate (flat)

Same steps in a slab, x from the salt side (x = 0) to the outer side (x = t = 5 mm):

$$k\frac{d^2T}{dx^2} + q_c = 0 \;\Rightarrow\; k\frac{dT}{dx} = -q_c x + A \;\Rightarrow\; T = -\frac{q_c x^2}{2k} + \frac{A}{k}x + B \tag{12}$$

$$q(0) = -k\frac{dT}{dx}(0) = -A = q_i, \qquad T(t) = T_o \;\Rightarrow\; T_w - T_o = \frac{q_i\, t}{k} + \frac{q_c t^2}{2k} \tag{13}$$

so h_c = k/t = 4520 W/m²K and $\Delta T_{gen} = q_c t^2/(2k)$ = 9.8 K.

Gas outside instead of a thermocouple (alternative)

If the outer surface temperature is unknown and loses heat to a gas, $q_o = h_g(T_o - T_a)$ with $h_g = h + h_{rad}$. Eliminating T_o with the energy check keeps the Robin form, with

$$H = \left(\frac{1}{h_c} + \frac{r_i}{r_o\, h_g}\right)^{-1}, \qquad T_{ref} = T_a + \Delta T_{gen} + \frac{q_c\left(r_o^2 - r_i^2\right)}{2\, r_o\, h_g} \tag{14}$$

4. How the Robin law becomes an OpenFOAM boundary condition

On the salt side the boundary cell (temperature T_c, its centre at distance $1/\delta$ from the wall, δ = deltaCoeffs) conducts the flux $k_s\,\delta\,(T_c - T_w)$ to the wall. Setting it equal to the Robin law ($H = h_c$, $T_{ref} = T_o + \Delta T_{gen}$ for the measured case):

$$k_s\,\delta\,(T_c - T_w) = H\,(T_w - T_{ref}) \tag{15}$$

Solving for the wall temperature gives a weighted average of the reference temperature and the cell temperature:

$$T_w = f\,T_{ref} + (1-f)\,T_c, \qquad f = \frac{H}{H + k_s\,\delta} \tag{16}$$

This is exactly OpenFOAM's mixed condition: refValue = T_{ref}, refGradient = 0, valueFraction = f. Limits: $H \to \infty$ gives a fixed wall temperature $T_w = T_{ref}$; $H \to 0$ gives an adiabatic wall. The coded condition re-evaluates H and T_{ref} every iteration, face by face (k(T) at the mean wall temperature, $T_o(z)$ at the face height).

5. Inner ring (mode ring, case A)

Radii: inner surface r_a = 1.25 mm (closed gas hole with the axis thermocouple), salt side r_b = 2.5 mm. The first condition is that no heat leaves through the hole:

$$q(r_a) = 0 \quad \Rightarrow \quad \frac{q_c\,r_a}{2} - \frac{A}{r_a} = 0 \quad \Rightarrow \quad A = \frac{q_c\,r_a^2}{2} \tag{17}$$

Heat flux into the salt at r_b, from (6):

$$q(r_b) = \frac{q_c\,r_b}{2} - \frac{q_c\,r_a^2}{2\,r_b} = \frac{q_c\,(r_b^2 - r_a^2)}{2\,r_b} \tag{18}$$

= 16.6 kW/m² (19.6 W over 75 mm). It contains **no temperature**: it is plain energy conservation, everything the ring generates, $\pi(r_b^2 - r_a^2)\,q_c$ per metre, leaves through its only exit, the circumference $2\pi r_b$.

Temperature drop across the ring, from (5) at r_a and r_b:

$$T(r_a) - T(r_b) = \frac{q_c}{4k}\left(r_b^2 - r_a^2\right) - \frac{q_c\,r_a^2}{2k}\ln\frac{r_b}{r_a} \tag{19}$$

= 0.92 − 0.42 = 0.49 K: the axis thermocouple (reading the ring's inner surface) is 0.49 K above the salt-side wall.

Only one more condition can be imposed. With $q(r_a) = 0$ one constant is used; one further condition fixes everything, so the flux into the salt and the wall temperature cannot both be prescribed (the salt's energy equation takes one condition per boundary point):

Choice	Imposed on the salt	Comes out of the CFD
flux (default ring)	q = 16.6 kW/m² into the salt (Neumann)	wall temperature → predicted thermocouple = T_w + 0.49 K: a genuine validation
temperature (impose_tc)	$T_w = T_{TC}(z) - 0.49$ K (Dirichlet)	flux into the salt: physically it must be 16.6 kW/m²; a difference shows missing physics

6. The difference between the two

	Outer cladding	Inner ring
What the wall is	pass-through: salt heat in, out to the measured outer surface	dead end: closed inside, nothing passes through
Known on the far side	temperature $T_o(z)$ (thermocouples)	zero heat flux (closed hole)
Flux at the salt	unknown, solved together with the salt	fixed by the ring's own heat: 16.6 kW/m² into the salt
Condition on the salt	Robin / mixed: $q_i = h_c(T_w - T_o - \Delta T_{gen})$, $h_c \approx 7.7$ kW/m²K	Neumann (fixed flux), or Dirichlet with impose_tc
The metal's own heat	raises T_w by ΔT_{gen} (6–10 K) and leaves to the outside	is the flux into the salt
Thermocouples	boundary data (imposed)	validation (+0.49 K), or imposed instead of the flux
OpenFOAM	codedMixed (k(T), spline $T_o(z)$)	externalWallHeatFluxTemperature (power) or codedFixedValue

In one sentence: the cladding **connects** the salt to a known temperature, so the heat it carries adjusts to the temperatures (it behaves like a thermal resistance); the ring connects the salt to **nothing**, so the heat it delivers is set by its generation and its temperature simply follows the salt next to it.